

For the Citrus Orchard there are three main objectives:

1. Distribute the year round excess citrus harvest to those who need it
2. Optimize structure of the trees through pruning, targeted fertilization, and mulching
3. Create an educational space around the benefits of pollinators plants around crops

Map:



1. Food Distribution

Background

Around 48 percent of undergraduate and 31 percent of graduate students face food insecurity at UC Santa Barbara. This is significantly higher than the 21 percent of Santa Barbara County residents who experience food insecurity. Any amount of food insecurity is an issue we must address but these statistics highlight the food insecurity of local college students. In addition to general food insecurity, many college students cannot afford to buy the most nutritious nor organic food. I have personally heard from the majority of my friends that they wish they could eat more organic and ethically sourced food but struggle to do so due to the marked up price. With the excess citrus produced in the citrus orchard, we can help reduce local food insecurity for college students and provide more nutritious and tasty produce. When I have shared the farm oranges with peers they comment that it is the best oranges they have ever eaten. With education and signage in the food bank, donating the fruit will not only aid in addressing food needs but also provide education on how locally and regeneratively grown produce is better for the environment and our taste buds!

Goals

The goals of this project are as follows: get approval for donating citrus fruits to the UCSB Food Bank, create and implement a plan for donating the fruit weekly, create educational signage about the benefits of locally grown produce, and reduce food waste on the farm. In order to get approval for donating to the food bank I am going to email the UCSB Food Bank coordinator, crystalb@as.ucsb.edu, to discuss donating the fruit over email or schedule an in-person meeting if possible. I will provide evidence of our safe and healthy growing practice

but make sure that we are not legally responsible for any effects just in case. Of course the fruit will be inspected for safe quality before donating. Next I will create a plan for distributing the produce which I will go into more detail in the Methods section. Thirdly, I would like to create educational signage, probably just printed google doc papers for replicability, to educate the student body on the benefits of locally and regeneratively grown produce. A positive byproduct of this project will be reducing food waste which will reduce smell and general waste in the citrus orchard and food stand.

Methods

To harvest the fruit the onsite team and interns can harvest the citrus throughout the week. On Friday I will collect the produce in one to two crates and drive to the UCSB food bank and donate it before it closes at 3 pm. Based on the amount of produce that is needed from the food bank, I can either increase or decrease the amount of crates that I bring. I will be in frequent communication with Becca and Stacy about the produce that is needed, how much is being donated, and other important information.

Equipment

Two crates for carrying the produce - 2 x \$15

Printed out paper for educational signage

Can be printed out at the Ellwood office or at the UCSB library

The overall cost of equipment is \$30

Timeframe/ Future

This project will begin as soon as possible in the Winter quarter of 2025, with the first donations occurring once approval has been granted from the UCSB Food Bank coordinator. I will continue to oversee this project through the Spring quarter of 2025. After that any other intern may continue this process by following this document.

Sources

Staff. "Food Insecurity - Health Indicators - Santa Barbara County - Cottage Health." *Food Insecurity*, Cottage Health, 2019,
www.cottagehealth.org/population-health/community-health-needs-assessment/2019/health-indicators/food-insecurity/.

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2. Citrus Tree Optimal Pruning

Background

There are many benefits to pruning trees. It encourages new healthy growth and gets rid of older branches that have reduced production or are susceptible to disease and pests. The new growth is likely to boost production by having greater production compared to older branches. Through reducing the leafy matter at the top, pruning encourages stronger roots because it redirects its energy towards below ground resilience. Pruning also reduces disease prevalence by either getting rid of diseased branches or promoting air flow throughout the tree which prevents diseases from spreading from branch to branch. Harvests are made more optimal when the height of trees is kept lower. Finally, pruned trees keep an orchard looking more tidy and aesthetic by sustaining healthier trees.

Goals

The goals of this project are to maintain the citrus orchard trees, boost quality production, make harvesting easier, and try out different pruning techniques and shapes. Pruning the citrus trees will ensure general upkeep by taking care of the health of each tree and encouraging new healthy growth. Production can be boosted through new growth but whenever a plant is pruned down and kept smaller, reduced production may occur. This may result in an equal or slightly less production than what is currently being produced. Nevertheless, the amount of citrus harvested due to time and labor will likely remain the same and there will just be less excess fruit left on the tree to rot. Furthermore, when the trees are kept smaller, harvesting is made more accessible and easier. The final goal of this project is to try out slightly varying styles of pruning to see how that impacts production and quality and size of the fruit.

Methods

For all the non experimental groups fertilizer can be applied if it would benefit the trees health.

This could include a general all purpose fertilizer, worm tea, or a liquid fertilizer high in Iron since that is one of the most common deficiencies. Adding mulch around existing trees will help conserve moisture for the trees and add organic matter and nutrients to the soil over time.

Furthermore, integrated pest management (IPM) should be employed to ensure the health of the trees against pests. Executing IPM requires close monitoring and precise actions against specific pests in a way that attempts to avoid the use of pesticides. Having native plants around the citrus orchard to attract predatory insects will help with pest control. This guide does a good job walking one through the process of IPM:

<https://ncipmhort.cfans.umn.edu/sites/ncipmhort.cfans.umn.edu/files/2022-03/2020-Guide-to-Integrated-Pest-Management.pdf>

There will be three different types of pruning methods done to either three to six trees. Ideally this experiment will be done to six trees for more replicable results, but the amount of trees treated depends on what the Pulice's are okay with. Besides these three to six test trees the rest of the trees will receive general pruning.

Control group: No pruning will be done to the trees. Harvesting can still occur.

Treatment group #1 Moderate Pruning: Reducing the height and size. Pruning out watershoots (suckers), old dead growth, crossing branches, and moderate pruning of inner branches for airflow.

Treatment group #2 Above Moderate Pruning: reducing the height and size to a greater extent. Pruning out watershoots (suckers), old dead growth, crossing branches, above moderate pruning of inner branches for airflow.

This video does a good job explaining fertilizer needs, pests, and pruning

<https://www.youtube.com/watch?v=Knbd2ni0PI>

In general, the branches to always prune are the three D's: Dead, Diseased, or Damaged.

The general approach to pruning a citrus tree is to keep it bushy, reduce the height when needed, and remove water shoots below the grafting point.

Equipment

Safety approval and paperwork may be needed to use some of the equipment such as the ladder

Pruning shears

Pruning saw

Gloves

Protective eyewear

Ladder

A labeling tap or signage to indicate the varying pruning techniques

Timeframe

Pruning will be done in the Winter quarter of 2025. The trees will be monitored and evaluated as the seasons progress and recorded in another google doc. Ideally the trees should be pruned as soon as possible because winter is the optimal pruning time. By the end this project will show the difference in tree health and fruit production between non pruned and pruned trees.

Future

The trees can continue to be pruned in the manner explained in the methods section. The experiment can be performed another year to strengthen the validity of the findings.

Sources

Espiritu, Kevin. "How to Plant, Grow, & Care for Citrus Trees (COMPLETE GUIDE)." *YouTube*, YouTube, 7 Mar. 2024, www.youtube.com/watch?v=_KnBD2ni0PI.

Krischik, Dr. Vera, and Laurie Schneider. "Pruning Decision Trees." *GeeksforGeeks*, University of Minnesota, 10 Apr. 2024, www.geeksforgeeks.org/pruning-decision-trees/.

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Osuna, Israel. "How to Prune Your Citrus Trees." *Four Winds Growers*, 8 Dec. 2024, www.fourwindsgrowers.com/a/blog/how-to-prune-your-citrus-trees#:~:text=Any%20growth%20coming%20from%20the,%2C%20damaged%2C%20or%20diseased%20branches.

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3. Educational Space

Background:

A permaculture showcase is what El centro is missing, and is what would compliment all of our hard work on the ranch so nicely. What we do as permaculturists and why we do it should *not* be a secret. In fact, it should be explained thoroughly to anyone who will listen so that we can share knowledge and inspire others. For example, if I told someone about the principle in permaculture that “the yield of the system is theoretically unlimited,” and explained that this essentially means that within a permacultural system, whether it be a tiny garden, or a large-scale farm, you can maximize the amount of food and other desired biomass by using every ounce of sunlight and rainfall efficiently. In turn, you can create unimaginable amounts of food, and in doing so you can even restore natural habitats.

This simple idea that food systems don’t have to look and be the way they are right now, is not something that people know. However, they should. Knowing where your food comes from and even how to grow it yourself is empowering. This space is a perfect place to empower others with knowledge on how to take care of themselves by taking care of the environment around them.

Goals:

The purpose of this space is to educate people who come through it on what is happening and why. As well as to showcase the following principles in action:

Principles of Permaculture

- Work with nature, rather than against the natural elements, forces, pressures, processes, agencies, and evolutions, so that we assist rather than impede natural developments.
- The problem is the solution; everything works both ways. It is only how we see things that makes them advantageous or not (if the wind blows cold, let us use both its strength and its coolness to advantage).
- Make the least change for the greatest possible effect.
- Everything gardens, or has an effect on its environment.
- The Yield of a System is Theoretically Unlimited: the only limit is the knowledge, information, imagination, and creativity of the designer.
- Relative Location: Elements in a system are viewed, not in isolation, but for the multitude of functional interconnections that they can have with the other elements of the design to enhance harmony.
- Each Element Performs Many Functions: By stacking functions, the designer has the forethought against the failure of one or more elements.
- Each Function is Supported by Many Elements: Maximizing beneficial connections between elements creates stability.
- Energy Efficient Planning: Through thoughtful design, we can make the most from the least. (zone planning, sector planning, slope)
- Use Biological Resources: By including a plant or animal in our design, we can increase our opportunities to save energy and increase yield.
- Energy Cycling: Each cyclical opportunity in the system increases the opportunity for yield.
- Small-Scale Intensive Systems: It's all about scale. Smaller systems are easier to respond to.
- Accelerating Succession & Evolution: Natural ecosystems develop and change over time. By observing these systems, we can design for effective restoration and productivity.
- Diversity: Functional relationships between elements creates stability and design opportunities.

From <https://www.freepermaculture.com/principles/>

Examples on how to showcase these principles will be named later.

Another goal of this project would be to extend and further incorporate the existing pollinator garden. One of the main ways that this space will be educational is explaining fruit tree guilds and how they are such a fundamental concept in permaculture. Fruit tree guilds are 3+ plants planted near each other (one being a fruit tree), that all benefit each other. Here is an example of a fruit tree guild I planted in my highschool:

Play

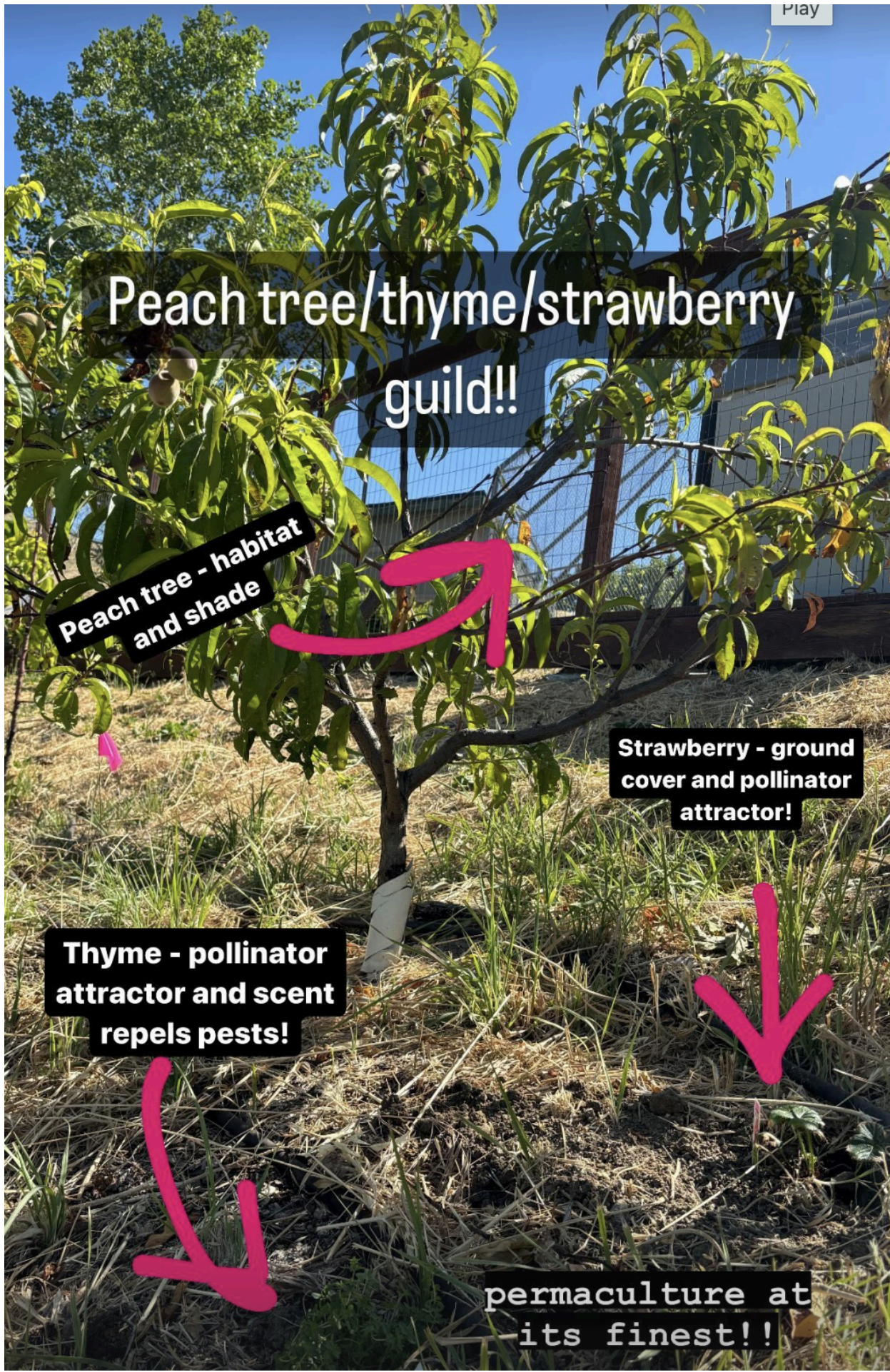
Peach tree/thyme/strawberry guild!!

Peach tree - habitat
and shade

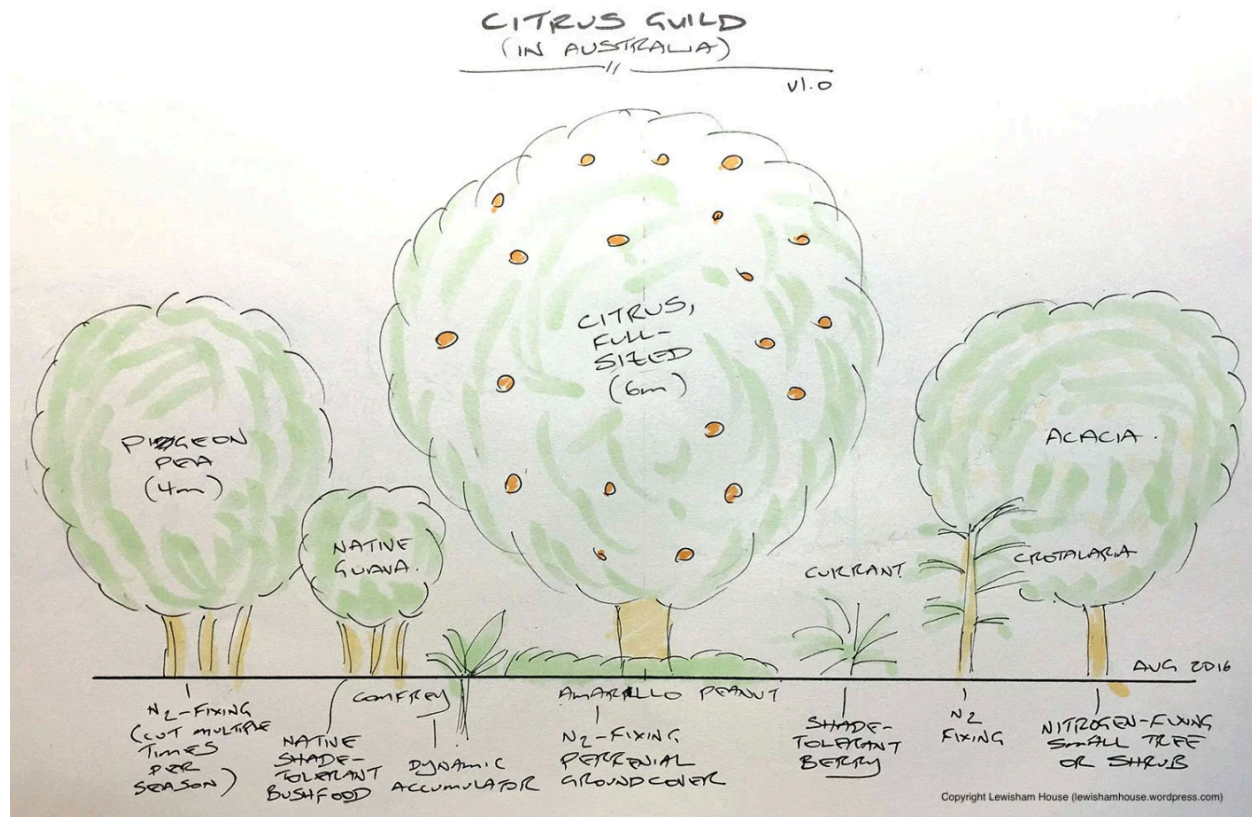
Strawberry - ground
cover and pollinator
attractor!

Thyme - pollinator
attractor and scent
repels pests!

permaculture at
its finest!!



Here is an example of citrus tree guilds:



- Notice that on the ground level, comfrey is planted as a **dynamic accumulator**, meaning that it takes nutrients from the soil and stores it in “tissues,” that are more bioavailable when the accumulator is harvested. This is similar to the ideas behind cover cropping. The plant acts as a natural fertilizer.
- Notice also that **shade tolerant** plants are planted at the base of the trees to make use of ALL the land, another very permaculture practice.
 - This also fits into ideas behind “layering” and food forests. Here is an amazing agroforestry/food forests video that talks a lot about layering and maximizing resources: <https://www.youtube.com/watch?v=KI3haUOkP-I>

- Notice lastly the nitrogen fixing plants planted down below, this is important because “Citrus trees are gross feeders. So the guild has to produce a lot of nitrogen, and other fertilisers, to support the citrus.” Gross feeders just means that the citrus need lots of nutrients and “food” to be healthy. There are a lot of synthetic citrus tree fertilizers, but fertilizers are anti-permaculture for a lot of reasons that will not be explored here. The point is, there is a way to produce all those fertilizers naturally, and that is what we will showcase.

Signage:

Since this is an educational space, signage is SUPER important. One example of signage in a permaculture garden is from the UMass Amherst Berkshire garden

(<https://www.localumass.com/permaculture/our-gardens/berkshire-garden/>).

Here are some pictures:



PERMACULTURE PRINCIPLE #10

Stack Functions

This Seaberry shrub provides a wealth of yields for us. Not only does it yield a nutrient-rich and medicinal berry, this plant also feeds the plants around it by fixing atmospheric Nitrogen and building soil. If that weren't enough, it also provides great habitat for songbirds and other flying critters.

Sea Berry

Hippophae rhamnoides
white fruit, orange flesh





Citrus tree guild with the blood orange tree:

- Goals: attract ladybugs, wasps and other predator insects to keep the aphids in check
- Break up the soil
- Accumulate nutrients in the soil

1) Soil builder - Yarrow



- California native
- Miwok tribe used the plant to make tea - used for headache/cold relief
 - Can also be used as a mouthwash for root care
- Drought tolerant
- Recommended to grow from seed!
- Perennial
- Deep roots - can help break up the soil (WithOUT tilling!)

- Insect attractor (specifically fungus-feeding lady beetle, a common lady bug in socal!)

2) NATURAL Pest repellent: dill + marigold

- Ladybug attractors (pollen rich)
- Marigolds strong scent also repels aphids
- Increases biodiversity
- Dill attracts hummingbirds
- Can grow BOTH from seed

Plant needs:

- drip line irrigation - can this be extended from the drip line that is in the pollinator garden right now? (ask josh)

Note: Nasturtium is a great biomass producer for the compost that we will eventually need to put into the citrus garden! Don't weed out the existing Nasturtium that is there! It also is a great soil cover.

Pollinator garden central piece:

Bees need water! To attract bees to this pollinator garden, we are going to give them water.

They like dirty water!

"It is well known that honeybees tend to collect water from many undesirable sources, such as rain-water gutters that are choked with decaying organic matter, on the puddles that form on the top of cow dung and sewage effluent, rather than from a source of clean water provided in the apiary for their use."

This [one is on etsy for \\$38.](#)





Put marbles and corks in the watering station for bees to stand on so that they won't drown.

Materials and initial cost:

- ☐ Seeds needed: yarrow, dill, marigold, one pack each (4-5\$ each)
- ☐ Fountain - ranging from 30-60\$
- ☐ Marbles and corks: \$15

Total INITIAL cost (before education signs are added after plants are established):

\$57-90

Sources:

<https://lewishamhouse.wordpress.com/2016/11/30/permaculture-citrus-guild-australia/>

<https://freckledcalifornian.com/2020/06/26/why-i-plant-and-grow-yarrow/>

<https://www.permies.com/t/55729/Lemon-tree-guild>

<https://myplantin.com/blog/dill-companion-plants#:~:text=Marigold,bring%20down%20the%20nuisance%20creatures.>

<https://backyardbeekeeping.iamcountryside.com/health-pests/creating-the-best-water-sources-for-bees/>